

12 Energy and respiration

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Glycolysis in cytoplasm converts glucose to pyruvate with a small ATP yield; link reaction and Krebs cycle feed reduced coenzymes into electron transport. Oxidative phosphorylation uses a proton gradient across inner mitochondrial membrane to drive ATP synthase. Oxygen is final electron acceptor. Anaerobic pathways regenerate NAD so glycolysis can continue.

Worked example: follow the reasoning

If oxygen is removed, electron transport stops, NADH is not efficiently reoxidised and oxidative ATP production falls sharply.

Quick check · multiple choice

A What is oxygen's role in aerobic respiration?

[1]

A First glycolysis substrate

B Only carbon source

C ATP synthase motor

D Final electron acceptor

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why cyanide blocking electron transport rapidly lowers ATP production despite available glucose.

[3]

Answers and explanation

Quick check: D. Final electron acceptor

Question 1 [3 marks]

Electrons cannot flow to oxygen [1]; proton gradient is not maintained [1], so ATP synthase makes far less ATP; only glycolysis remains [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.